Literature Status: Hereditary Counts of Spreading Models

Run `fa_banach_001`

2026-06-23

Status. Literature already answered. This packet records that several hereditary spreading-model questions from Androulakis–Odell–Schlumprecht– Tomczak-Jaegermann, *On the structure of the spreading models of a Banach space*, arXiv:math/0305082, are explicitly answered in later literature.

Source questions. The source questions are TeX labels `qst3.3` and `qst3.5` in the arXiv source, printed as Questions 3.15 and 3.17 on pages 19–20 of the source PDF. They ask, among other things:

- whether a Banach space whose every infinite-dimensional subspace admits spreading models equivalent to the unit vector bases of ℓ_1 and ℓ_2 must contain more, perhaps uncountably many, mutually non-equivalent spreading models;
- whether, for each $n \in \mathbb{N}$, there is a Banach space whose every subspace has exactly n different spreading models;
- whether there is a Banach space whose every subspace has exactly countably many different spreading models.

Supporting answer. Beanland–Freeman–Motakis, *The stabilized set of p's in Krivine's theorem can be disconnected*, arXiv:1408.0265, gives the required examples. Its Theorem 1, stated on page 2 of the supporting PDF, says that if

$$F \subseteq [1, \infty]$$

is either finite or an increasing sequence together with its limit, then there is a reflexive Banach space X with an unconditional basis such that every infinite-dimensional block subspace Y has the following behavior:

- ℓ_p is finitely block represented in Y exactly for $p \in F$;
- if F is finite, then the spreading models admitted by Y are exactly the spaces ℓ_p, $p \in F$;
- if F is an increasing sequence with limit p_ω, then every spreading model admitted by Y is isomorphic to ℓ_p for some $p \in F$, and every non-limit $p \in F$ is admitted.

On page 3, the same paper explicitly says that Theorem 1 solves the AOST questions about exactly n many spreading models, exactly countably infinitely many spreading models, and whether hereditary ℓ_1 and ℓ_2 spreading models force uncountably many spreading models.

Consequences for the AOST questions. Taking F finite with $|F| = n$ gives the exactly-n examples. Taking F to be an increasing sequence together with its limit gives a countable family of possible ℓ_p spreading models, and the supporting authors identify this as the solution to the countably-infinite question. Finally, taking $F = \{1, 2\}$ gives a space whose hereditary subspace structure admits ℓ_1 and ℓ_2 spreading models but not uncountably many different ones, answering the uncountability alternative negatively.

Provenance. This is an explicit literature answer: Beanland–Freeman–Motakis cite AOST and state that their Theorem 1 solves these open questions. The packet is therefore not an original proof or a new partial result.

Scope limitations. This packet covers only the hereditary counting questions above. It does not address AOST's other questions about stabilizing all spreading models, the realisable partially ordered structures of $SP_\omega(X)$, consequences of finite or countable $SP_\omega(X)$, or the unique-spreading-model question.

References

[1] G. Androulakis, E. Odell, Th. Schlumprecht, and N. Tomczak-Jaegermann, *On the structure of the spreading models of a Banach space*, arXiv:math/0305082; Canad. J. Math. 57 (2005), no. 4, 673–707.

[2] K. Beanland, D. Freeman, and P. Motakis, *The stabilized set of p's in Krivine's theorem can be disconnected*, arXiv:1408.0265.